

SliderMC Command Cheat Sheet

Firmware V1.0. Same groups as the [printable sheet](#). Canonical prose: [protocol.md](#). Dual-axis timing: [dual-movement.md](#). Joystick: [motion-joy.md](#). Working window: [working-window.md](#).

Regenerate: `python tools/render_command_cheatsheet.py`

Call — send this (brackets = optional; bare = omit args).

Reply — typical success line; `=` = silent (errors still `!E:...`).

Arguments

`<s>` : Speed in mm/s or deg/s

`<a>` : Acceleration / deceleration in mm/s² or deg/s²

`<axv>` : Axis values for the packed live channels (STEP/DIR motors first, then RC servos; up to six). A line is **either** positional **or** named — never mixed (MT 20 C100 → `!E:parse`). : **Positional**: one number per channel in order, spaces between them. `_` skips a slot and leaves that channel unchanged (MT `_ _ 100` moves only channel 3). Omitted trailing slots are unchanged (MT 50 is channel 1 only). : **Named**: letters X Y Z = motors 1–3, A B C = servos 1–3, then the value (spaces optional: C100, C 100, MT C100 B50). Duplicate letter, unknown letter, or a channel that is not fitted → `!E:parse`. : PD uses the same forms in `µm` (`<axv_µm>`); skip `_` or omit named → `0`. : MJ uses signed **percent** (not mm); omitted named extra = `0`; `_` is invalid.

S — Set (session)

COMMAND	CALL	REPLY	DESCRIPTION
Set Speed	SS [<code><v></code>]	—	Cruise speed mm/s ($\leq$ max_speed_1); bare reloads init_speed; live on next fill (incl. MJ). Dual MT: axis1=session, axis2×ratio.
Set Accel	SA [<code><a></code>]	—	Peak accel mm/s ² ($\leq$ max_accel_1); bare reloads init_accel; live on next fill (incl. MJ). Dual MT: same ratio scaling as SS.
Set Enable	SE [0 1]	—	Driver enable 0 1; bare toggles; required before motion; off = hard stop. SE 0 also stops servo PWM (limp).
Set Terminal	ST [0 1]	—	Terminal Mode 0 1; bare toggles; local echo + UART sniff to USB (expert).
Set Verbose	SV [0 1]	—	Verbose #... push 0 1; bare toggles; ~3 Hz (rate via verbose_rate_hz).
Set Debug	SD [0..5]	—	USB-only debug level 0.5; bare restores default; never on UIC UART.
Set Left	SL [<code><axv></code>]	—	Session soft min (working window); bare→MOTOR_N_min/SERVO envelope; none clears (→envelope if set); skip <code>_</code> ; <code>!E:limit</code> past envelope.
Set Right	SR [<code><axv></code>]	—	Session soft max; bare→MOTOR_N_max/SERVO envelope; none clears; skip <code>_</code> ; <code>!E:limit</code> if left>right.
Set Position	SP [<code><axv></code>]	—	Set reported pose (no motion); idle only; bare/0 = here is zero; skip <code>_</code> .

G — Get (session)

COMMAND	CALL	REPLY	DESCRIPTION
Get Speed	GS	GS: <mm/s>	Current session cruise speed.
Get Accel	GA	GA: <mm/s ² >	Current session acceleration.
Get Enable	GE	GE:0 1	Driver enable state.
Get Terminal	GT	GT:0 1	Terminal Mode state.
Get Verbose	GV	GV:0 1	Verbose push state.
Get Debug	GD	GD: <0..5>	USB debug level.
Get Left	GL	GL:<pos> [...]	Session soft min; effective (session else envelope); - if both None; extra pipe fields when packed≥2.
Get Right	GR	GR:<pos> [...]	Session soft max; same effective / - rules as GL.

I — Is / Info

COMMAND	CALL	REPLY	DESCRIPTION
Is Moving	IM	IM:0 1	Moving or settling on any active axis.
Is Homing	IH	IH:0 1	Homing cycle active.
Is Limit	IL	IL:0 1	At soft-limit position (axis1).
Is Error	IE	IE:0 1	PIN_DRV_ERROR / EMO latched.
Is Position	IP	IP:<pos> [...]	One pipe field per packed live channel (motors then servos; CG axis = sum).
Is Axis	IA	IA:<n>	Packed live channels motors+servos (1..6). Same as CG axis. CS axis rejected.
Is Target	IT	IT:<pos> -	Axis-1 seek target, or - if none / soft-stop.
Is Ready	IR	IR:0 1	1 only if idle, not homing, enabled, and not waiting.
Is Waiting	IW	IW:0 1	1 if any WT / WM / WH / WP / WC / WN wait is active.
Is Diag	ID	ID:underrun=N peak_hz=... overshoot=... fifo_min=...	Motion diag counters (FIFO underrun, peak STEP Hz, ...).
Is Cause	IC	IC:<reason>	Last chip reset: power wdt run soft debug brownout ...
Is GPIO	IG	(multi-line table)	ASCII GP / name / desc ($\leq$ 80 cols). Extra-axis rows only if that axis is live.

M — Movement

COMMAND	CALL	REPLY	DESCRIPTION
Move To	MT <axv> (absolute)	—	Absolute user units; up to 6 packed channels (XYZ motors, ABC servos); skip <code>_</code> ; needs SE; live-retarget. Dual MT: time-sync. Out-of-window = <code>!E:soft</code> (no clip).
Move By	MB <axv> (delta) ...	—	Relative move; same skip/named XYZABC rules as MT.
Move Joy	MJ <pct> ...	—	Joy speed % of SS, signed (– left / + right); omit named extra=0; 0=soft-stop; SS/SA live; clamp max_speed_N. Hold-to-jog: MJ ±100, MS on release.
Move Home	MH [1 2 3]	—	Homing; axis 1 (default), 2, or 3; no-op if home_mode_N=0; cancel MS/ME.
Move Soft Stop	MS	—	Soft decelerate both axes; keeps enable; ends joy-mode; does not cancel waits. Dual: scaled accel kept.
Move E-Stop, Halt	ME	—	Immediate STEP abort; enable off; cancel waits and remaining ; chain.

P — Path			
COMMAND	CALL	REPLY	DESCRIPTION
Path Clear	PC	—	Clear path buffer (count→0); !E:busy while PG active.
Path Data	PD <axv_um>	—	Append signed μm sample(s); up to 6 packed channels; skip _ →0; OK while PG (live stream).
Path Go	PG	—	Play buffer from sample 0; needs SE; !E:empty busy disabled. MS/ME ends path.
Path Number	PN	PN: <count>	Samples in buffer; allowed during PG.
Path Slice	PS [<us>]	—	Slice length μs (≥1000); bare→init_path_slice_us; !E:busy while PG.

W — Wait			
COMMAND	CALL	REPLY	DESCRIPTION
Wait Time	WT [<sec>]	—	Delay then continue ; chain; bare→1 s; never !E:timeout.
Wait Moving	WM [<timeout_s>]	—	Pause chain until move ends; optional timeout → !E:timeout, cancel rest of chain.
Wait Homing	WH [<timeout_s>]	—	Pause until homing ends; timeout same as WM.
Wait Pos	WP <pos> [<timeout_s>]	—	Wait until time-sync master pos reached/overstepped; idle→immediate; 2nd arg=timeout.
Wait Cruise	WC [<timeout_s>]	—	Wait until cruise (status M) or idle; optional timeout → !E:timeout.
Wait Not cruise	WN [<timeout_s>]	—	Wait until not cruise M; idle/A/B→immediate; timeout same as WM.

E — Extender			
COMMAND	CALL	REPLY	DESCRIPTION
Ext Out	EO 0..3 [0 1]	—	Ext out n logical 0 1; bare EO0 toggles; glued EO01≡EO0 1; OK during EMO. EO4+ rejected.

Special			
COMMAND	CALL	REPLY	DESCRIPTION
Reboot	RB	—	Soft MCU reset (no power cycle); EN off first. CS motors/servos do not need RB.
Beep	BE [<ms>]	—	Pulse PIN_BUZZER; bare 100 ms; clamp 1..1000; not a wait. No-op if BUZZER_use=0. OK during EMO/path.
Camera Trigger	CT [<ms>]	—	Pulse PIN_CAMERA_CTRL low (OC sink); bare 100 ms; clamp 1..60000; not a wait; motion continues. OK during EMO/path. Verbose on: status letter T while pulsing (at least once).
Help	HL/? HL ?	(multi-line table)	Two-column table of all commands (short + phrase). ? needs newline; not realtime status.

C — Config			
COMMAND	CALL	REPLY	DESCRIPTION
Config Set	CS <key> <value>	—	Persist key to mc.ini; silent ok. CS motors/servos re-init GPIO immediately (no RB). CS axis rejected.
Config Reset	CR	—	Reset all config to compiled defaults and save mc.ini.
Config Get	CG [<key>]	CG: <key>= <value>	One key, or bare dumps all keys (multi-line).

V — Version			
COMMAND	CALL	REPLY	DESCRIPTION
Version About	VA	VA:...	About string (name, version, author).
Version FW	VF	VF:<version>	Firmware version.
Version Protocol	VP	VP:1	Protocol version (1). Wire may change without bumping VP.
Version GPIO	VG	VG:PIN_*=n (multi-line)	Machine-readable pin map. Extra-axis pins if that axis is live.

Now Commands

Executed immediately without \n

COMMAND	CALL	REPLY	DESCRIPTION
Status now	#	#<state> ...	Realtime compact status line (no newline).
Soft Stop now	!	—	Realtime soft stop (same urgency as MS).
E-Stop now	ESC	—	Realtime halt (same urgency as ME): STEP abort, EN off, cancel waits/chain.

Notes

- Chain with ; / comments from that character to end of line. Realtime (no newline): # status, ! soft stop, ESC halt, Ctrl-X soft reset. # is not a comment. ? is help (needs \n), not status.
- Path mode (PG): most move/session cmds → !E:busy; allowed: MS/ME/RB/PD/PN/I*/G*/V*/IG/HL/?/CG/BE/CT.
- MJ / Move Joy: signed % of SS; skip unchanged values; SS/SA live in joy-mode. Hold-to-jog: SS then MJ ±100, MS on release. See [motion-joy.md](#).
- Skip token _ only (MT/MB/PD/SL/SR). Named X/Y/Z/A/B/C is an alternative (not mixed with positional). SL/SR none clears a side (effective = envelope when set). See [working-window.md](#).
- Envelopes / units: MOTOR_N_min/MOTOR_N_max, SERVO_N_min/SERVO_N_max, synthesized axis_min_N, steps_per_unit_N, unit_name. Servo pulse SERVO_N_min_pulse/max_pulse (default 500–2500 μs; analog 1000–2000); SERVO_N_swap reverses sense. CS axis and CS slider_* are rejected.
- Breaking wire (ME, ? help, # status only, / comments, CT) does **not** bump VP — still VP:1.